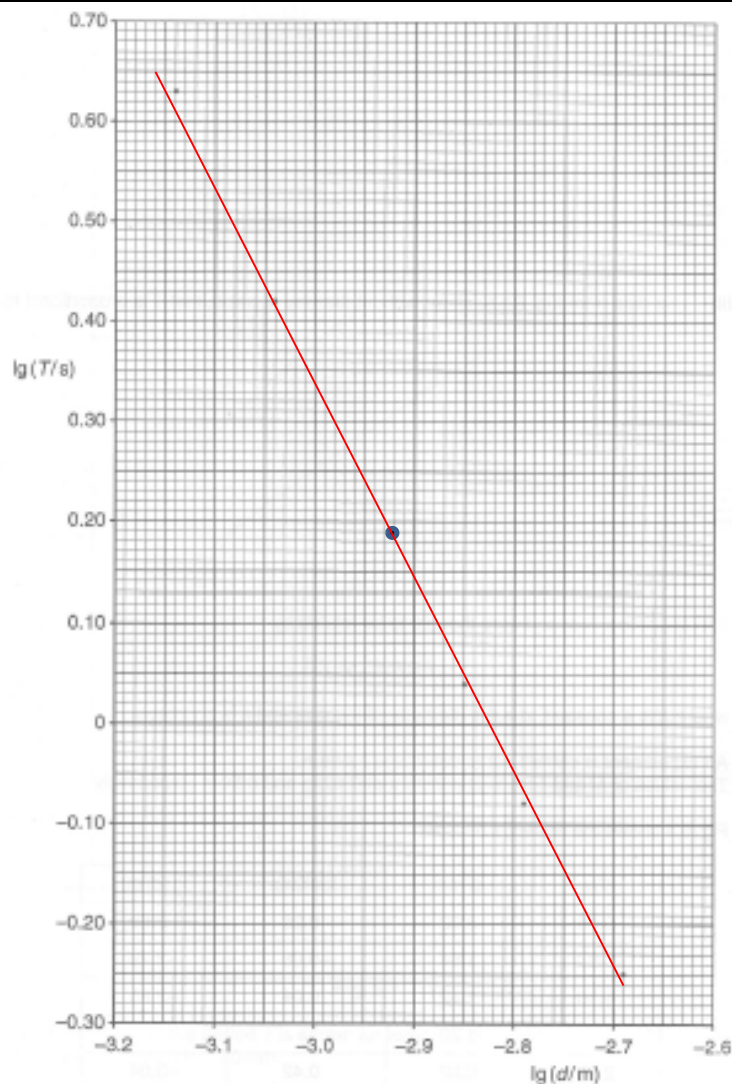


6(a)(i)	The period T increase with increase in l but with a decreasing rate.
6(a)(ii)	T is proportional to $\sqrt{M}$ implies $\frac{T}{\sqrt{M}} = \text{constant for a fixed } l$. From Fig. 6.3, for $l = 1.0 \text{ m}$, at $T=1.8 \text{ s}$, $\frac{T}{\sqrt{M}} = \frac{1.8}{\sqrt{0.265}} = 3.5$ at $T=2.1 \text{ s}$, $\frac{T}{\sqrt{M}} = \frac{2.1}{\sqrt{0.365}} = 3.5$ at $T=2.4 \text{ s}$, $\frac{T}{\sqrt{M}} = \frac{2.4}{\sqrt{0.475}} = 3.5$ Since the ratio $\frac{T}{\sqrt{M}} = 3.5$ is constant, T is proportional to $\sqrt{M}$ (shown). The majority of answers provided three or more values for the constant of proportionality. Some answers were left as a fraction instead of calculated to two significant figures.
6(a)(iii)	T is proportional to $\sqrt{l}$ implies $\frac{T}{\sqrt{l}} = \text{constant for a fixed } M$. From Fig. 6.3, for $M = 0.2 \text{ kg}$, at $l = 0.25 \text{ m}$, $\frac{T}{\sqrt{l}} = \frac{0.78}{\sqrt{0.25}} = 1.56$ at $l = 0.50 \text{ m}$, $\frac{T}{\sqrt{l}} = \frac{1.1}{\sqrt{0.50}} = 1.56$ at $l = 1.00 \text{ m}$, $\frac{T}{\sqrt{l}} = \frac{1.56}{\sqrt{1.00}} = 1.56$ Since the ratio $\frac{T}{\sqrt{l}} = 1.56$ is constant, T is proportional to $\sqrt{l}$ (shown).
6(b)(i)	For $d = 1.20 \text{ mm}$, $T = 1.56 \text{ s}$, $\lg T = 0.193$, $\lg d = - 2.92$

6(b)(ii)



A significant number of best-fit lines were incorrectly drawn, with points distributed unevenly about the line.

6(b)(iii)

$$\text{gradient} = \frac{0.53 - (-0.24)}{-3.1 - (-2.7)} = -1.93$$

6(c)(i)

Log both side of the equation given in (b),

$$\lg T = \lg \left[K (Ml)^{1/2} \right] + n \lg d$$

Thus, n is the gradient of the graph.

6(c)(ii)

$n = -2$ since n is an integer.

6(d)	From Fig. 6.5, when $\lg d = -2.7$, $\lg T = -0.24$. Thus, $\lg T = \lg \left[K (Ml)^{1/2} \right] + n \lg d$ $\Rightarrow -0.24 = \lg \left[K (0.40 \times 0.50)^{1/2} \right] + (-2)(-2.7)$ $K = 5.12 \times 10^{-6} \text{ SI unit}$
6(e)(i)	Measure period T of oscillations and M , mass of the rod.
6(e)(ii)	The setup is that of a torsional pendulum. If the oscillations are simple harmonic, the period of the torsional pendulum is given by $T = 2\pi \sqrt{\frac{I}{k}}$ where I is the moment of inertia of the rod and k is the force constant of the wire. Moment of inertia of the rod is given by $I = \frac{ML^2}{12}$ where M is the mass of the rod and L is the length of the rod. So for a fixed length, varies M and measure T. Plotting T^2 against M, the theory is supported if a straight line is obtained.